

Sravan Sai Methuku

Data Scientist

Saint Louis, MO | +1 (314) 312-1223 | sravansaimethuku45@gmail.com | [Linkedin](#) | [Github](#)

SUMMARY

Data Scientist with 4+ years of cross-industry experience driving measurable ROI boosted SKU-level demand forecast accuracy 20%, reduced fraud losses by \$250K/Quarter, and saved \$1.2M in operational costs. Adept at designing end-to-end ML workflows data preprocessing, feature engineering, model selection, hyperparameter tuning, and CI/CD-driven deployment of production grade pipelines. Specialize in NLP and LLM fine-tuning alongside explainable AI techniques to ensure transparency and trust. Proven ability to translate complex data insights into actionable, customer-centric strategies through close collaboration with cross functional teams.

TECHNICAL SKILLS

Languages & Querying | Python (Pandas, NumPy, PySpark)

SQL (PostgreSQL, Redshift), R

Machine Learning & Deep Learning |

Algorithms: Transformers, CNN, RNN, LSTM, GAN,XgBoost, Random Forest, SVM, Decision Tree, Linear Regression , K-Nearest Neighbors, SVM, Naive Bayes,

Model Development | PyTorch,TensorFlow, scikit-learn MATLAB (Deep Learning, Statistics, Computer Vision, Optimization, Parallel Computing, Signal Processing, Reinforcement Learning Toolboxes)

Genai tooling | Hugging Face Transformers, LangChain (RAG, PEFT/LoRA) OpenAI API , Prompt Layer

Mlops | Mlflow ,Feast Feature Store, GreatExpectations

&Open Lineage, Apache Airflow GitLab CI/ GitHub actions

Docker & Kubernetes Terraform, Evidently AI, fast Api

Cloud & Lake-house | AWS (S3, SageMaker, Glue, Kinesis), GCP(Vertex AI, Big Query), Azure (Synapse, Databricks)

Databases & Data Engineering | MySQL, PostgreSQL, MongoDB, Apache Spark, Databricks, Snowflake, Kafka, Teradata

Data Visualization | Power BI, Tableau , Streamlit, Plotly

Tools & Collaboration | Git, Linux/Unix, JIRA (Agile/Scrum)

Operating Systems | Windows, Linux, Mac OS, Unix

WORK EXPERIENCE

Anheuser-Busch InBev | Aug 2024 – Present | Data Scientist

- Built a demand forecasting pipeline using **PySpark**, **Spark SQL**, and models like **ARIMA** and **Gradient Boosted Trees** optimized with **grid search**, reducing **MAPE by 20%** and improving working capital efficiency by **15%**.
- Designed and Executed causal A/B tests and uplift modeling for **BEES B2B**; deployed **TensorFlow Recommenders & AutoML XGBoost** in **Databricks** Feature Store, achieving **12 %** basket lift & **20 %** cross-sell uplift across 200k retailers.
- Trained **Random Forest regression models** over **5 billion** IoT sensor records in **Azure Synapse / Spark mllib**, optimizing **feature selection** and **hyperparameter tuning**; delivered a **8 %** yield boost and **10 %** cost reduction for smart barley agronomy.
- Deployed a **machine learning microservice** on **Vertex AI** to optimize delivery time predictions, using real-time data from **Kafka** and **Kinesis**. Reduced ETA by 15% and logistics costs by **10%**, while maintaining same-day delivery SLAs.
- Developed a predictive maintenance system using **Convolutional Neural Networks (CNNs)** to analyze **SCADA equipment imagery**, improving **uptime by 45%** and saving **\$1.2M in operational costs**. Automated workflows with **Airflow**, **MLflow**, GitHub Actions, and **Terraform**, with model monitoring via **Evidently**.

Centene Corporation | Dec 2023 – May 2024 | Data Scientist / ML Engineer

- Engineered automated **ETL pipelines** with **Apache Airflow**, **PySpark**, & **AWS Lambda** for HIPAA-compliant healthcare data, improving data pipeline **efficiency by 60%** and ensuring secure ingestion and transformation.
- Built **supervised classification models** using **XGBoost**, **Random Forest**, and **scikit-learn** to predict 30-day hospital readmissions; applied **cross-validation**, **ROC-AUC**, and **feature engineering**, reducing readmission costs by **12%**.
- Applied **NLP techniques** with **spaCy**, **BERT**, and **TF-IDF** to extract clinical features from EHR notes and call transcripts, boosting **recall by 15%** and cutting manual chart reviews by **20%**.
- Developed real-time monitoring pipelines with **Kafka**, **Amazon Kinesis**, and **Redshift** to detect high-risk events like ER visits and medication triggers; enabled timely interventions and reduced escalations by **18%**.
- Designed interactive executive dashboards in **Power BI** and **Tableau**, tracking clinical KPIs, treatment adherence, and program outcomes across 15+ Medicaid programs.
- Designed end-to-end **ML pipelines** and **MLOps workflows** using **Docker**, **Kubernetes**, **Jenkins**, **Flask**, and **FastAPI**, reducing model release time by **50%** and maintaining **99%** uptime for care-management APIs.

Glenmark Pharmaceuticals, Bangalore, India | Sep 2022 – Aug 2023 | Data Scientist

- Automated **ETL pipelines** using **Apache Airflow, Talend, PostgreSQL**, reducing data load time by **40%**.
- Built **biostatistical ML models (Random Forest, lightgbm, Tensorflow)** with feature engineering, **hyperparameter tuning** And **SHAP**, boosting accuracy **30 %** and speeding go/no-go decisions.
- Built pharmacovigilance detection system (**SQL, Python, pandas, numpy, spacy NLP**) to process adverse event narratives
- Built dashboards in **Power BI** and **Tableau** to monitor clinical KPIs. Applied A/B testing and conversion analysis using **Python, Pandas**, and **Scikit-learn**. Improved **patient recruitment efficiency by 15%**.
- Deployed ML workflows (**Docker, Kubernetes, mlflow, Terraform**); ensured **99 %** availability and **50 %** faster releases.
- Utilized **statistical methods** (e.g., **ANOVA, chi-square tests, ROC-AUC**) and **data preprocessing** (e.g., **imputation, normalization**) to enhance modeling reliability and reproducibility.

American Express, Bangalore, India | March 2021 – Aug 2022 | Data Analyst/Junior Data Scientist

- Built fraud and risk dashboards using **Power BI, Tableau, Dax** driving real-time decision-making and reducing financial loss.
- **Built fraud detection and credit-risk models** using **Logistic Regression, Decision Trees**, and **XGBoost**; improved model precision by **22%** and helped save **\$250K** quarterly in fraud-related losses.
- Conducted **EDA & data modeling** with Python (**Pandas, pyspark**), **SQL (postgresql, Redshift)**, and **Azure Synapse** for
- **churn, segmentation, credit-risk** models; boosted **precision 18 %** and accelerated deployment via **github Actions CI/CD**.
- Performed customer segmentation and **credit-risk scoring** using PySpark; **streamlined ETL workflows** with dbt, SQL procedures, and GitHub Actions, boosting model **accuracy** by 18% and improving pipeline reliability.
- Engaged with fraud, risk, and data teams to deploy **analytics solutions** to enhance efficiency with strong **business impact**.

EDUCATION

Master of Science in Applied Computer Science	May 2025
Southeast Missouri State University, MO, USA	

CERTIFICATIONS

- AWS Certified Machine Learning – Associate
- Microsoft Certified: Azure Data Engineer Associate
- Python Data Structures – Coursera / University of Michigan